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Fundamental Study to Develop Bone-Mimicking Phantom

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Since the recent development of ultrasonic diagnosis of osteoporosis, ultrasonic examination of bones has received considerable attention. Because of the large attenuation in bone and large difference in the acoustic impedance between bone and adjacent tissues, ultrasonic diagnosis of bone is generally difficult to perform. The purpose of this study was to develop a bone-mimicking phantom for fundamental studies of ultrasonic diagnosis of bone. The phantom developed here was made of plaster. The acoustic characteristics on ultrasound of the developed phantom showed fairly good agreement with bovine cancellous bone, which is analogous to human bone.

KEYWORDS: bone-mimicking phantom, ultrasonic diagnosis, osteoporosis, bovine cancellous bone

Since ultrasonic diagnosis of bone is difficult, ultrasonic studies of this tissue are not common. The reasons for this difficulty are that the attenuation of the bone is large, the difference in impedance from adjacent tissues is also large and transfer of ultrasound into bone is not easy. In recent years, however, ultrasonic visualization and ultrasonic diagnosis of osteoporosis have been attempted.^{1–3)} Studies of this tissue should be encouraged for further development of ultrasonic diagnosis. The purpose of this study is to develop a bone-mimicking phantom, the acoustic characteristics of which are similar to those of real bone, for use in fundamental studies of the ultrasonic diagnosis of bone. The main material of the phantom developed here was plaster. The velocity and attenuation for ultrasound of the phantom showed good agreement with bovine cancellous bone which is analogous to human bone. Diagnostic images obtained from the phantom were fairly similar to those from bovine bone.

Cancellous bone consists of a trabecular structure filled with marrow.⁴⁾ Instead of human bone, we used bovine cancellous bone in this study. Figure 1 shows a schema of bovine thighbone. Because large and uniform specimens could be obtained, reference cancellous bone (40 × 40 × 10 mm) was excised from the portion indicated in the figure. The acoustic characteristics of cancellous bone depend on the structure, and the ratio of the volume of trabecular structure to that of marrow. Since the structure is complicated, we employed the density of the reference cancellous bone as an index of the ratio. The density calculated from the dimensions and the specific gravity was approximately 1.8 g/cm³. In order to obtain acoustic characteristics like those of cancellous bone, experimental construction of the phantom fabricated in this study proceeded as follows. Lime, the main ingredient of which, like bone, is Ca, CMC vegetable adhesive (Tokyo Sanhomu Co., Ltd.) and chopped flax (diameter: 0.4 mm, length: approximately 2 cm) were used. Flax was used in order to achieve a meshlike structure like that of cancellous bone and to adjust the attenuation. Figure 2 shows the fabrication procedure for the phantom. First, all the materials (lime, flax and CMC) are mixed thoroughly, poured into a mold and allowed to stand in a degassing chamber for one day. Next, the specimen is placed in a thermostatic chamber to dry and harden. The specimen is cut into pieces with the same dimensions as the reference cancellous bone described above. Finally, the specimen is immersed in degassed water. The density of the specimen was approximately 1.7 g/cm³, nearly equal to that of the reference bone. The ratio of flax to lime was defined

as the weight percent (wt%) of plaster in the total weight of the specimen (lime+flax+CMC). In the measurement of the acoustic characteristics of the specimen, this percentage was employed as a parameter.

Figure 3 shows a schematic diagram of the measuring system. The specimen is placed in a water tank filled with degassed water and irradiated with ultrasound using a function generator and plain transducers (10 mm diameter). The received waves are input to a personal computer to calculate the sound velocity and attenuation in the specimen. The water temperature in the tank can be maintained within ±0.3°C of the measuring temperature.

Attenuation α which does not include apparent attenuation with reflection at both sides of the specimen and the bone is calculated as

$$\alpha = \frac{-40 \log \left(\frac{1}{1 - R_p} \right) - 20 \log \left(\frac{1}{T_{p1} T_{p2}} \right) - 20 \log \left(\frac{V_1}{V_2} \right)}{d}, \quad (1)$$

where d : thickness, R_p : reflection coefficient, T_{p1} : transmis-

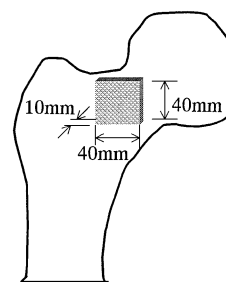


Fig. 1. Bovine thighbone and the portion where the reference cancellous bone was excised.

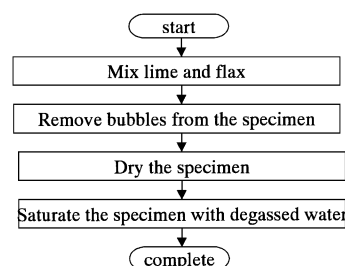


Fig. 2. Fabrication procedure for the phantom.

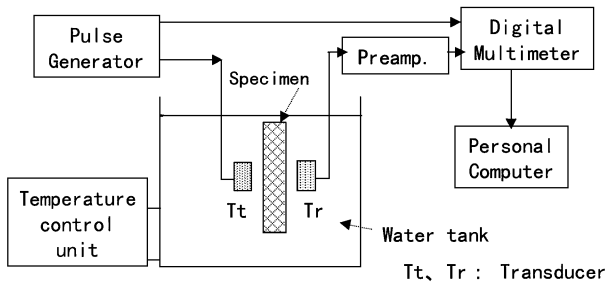


Fig. 3. Schematic diagram of measurement system.

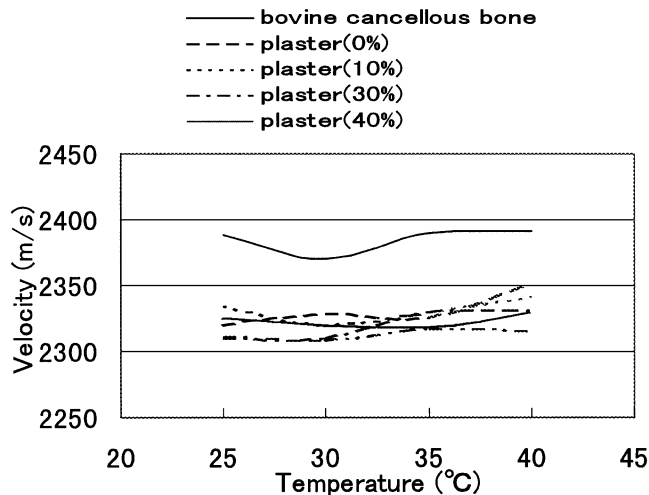


Fig. 4. Temperature dependence of the velocity for phantom specimens and bovine cancellous bone.

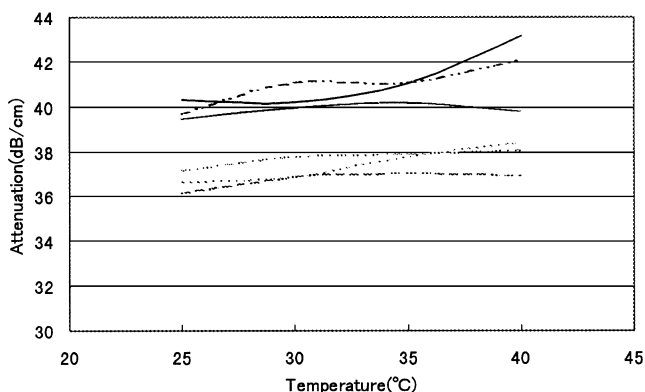


Fig. 5. Temperature dependence of the attenuation for phantom specimens and bovine cancellous bone.

sion coefficient at the front surface, T_{p2} : transmission coefficient at the back surface, V_1 : transmitted voltage and V_2 : received voltage.

Measurements were performed at various water temperatures and ultrasonic frequencies. The temperature was varied between 25°C and 40°C in 5° steps, and the frequency from 1 MHz to 10 MHz in 1 MHz steps.

Figure 4 shows the relationship between sound velocity and temperature. As shown, the sound velocities obtained from the specimens are between 2310 m/s and 2350 m/s, values similar to those of the reference bovine cancellous bone, between 2360 m/s and 2390 m/s.

Figure 5 shows the relationship between attenuation and temperature. Figure 6 shows the relationship between atten-

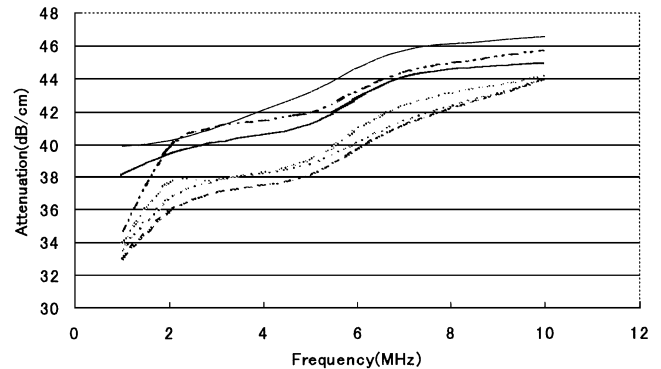


Fig. 6. Frequency dependence of the attenuation for phantom specimens and bovine cancellous bone.

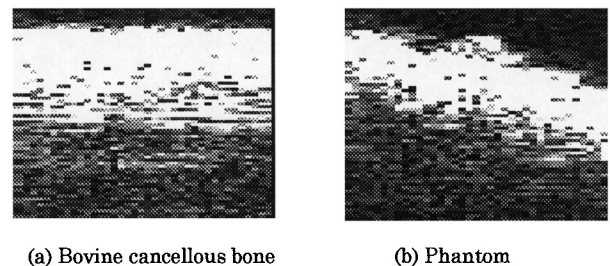


Fig. 7. Images obtained using ultrasonic diagnostic equipment.

uation and frequency at 35°C. From these measured data, it is clear that attenuation of the specimen is very similar to that of the reference bovine cancellous bone, especially at the ratio of flax to lime of 30%. The reason that the attenuation in the 40% flax to lime sample is small, is that the density of the specimen is decreased due to the loss of internal structure caused by the reduced adhesive CMC. In contrast, this reduction in attenuation due to the loss of internal structure shows the possibility of changing the attenuation by using varying amounts of an appropriate adhesive.

Figure 7 shows examples of the images of bone (a) and phantom (b) obtained using ultrasonic diagnostic equipment. The white patterns in (a) are the images of the echoes from the trabecular structures of the bone. Similar images are seen in (b).

In order to develop a bone-mimicking phantom for use in the development of ultrasonic diagnosis, a fundamental study was carried out. Bone phantoms, the main constituent of which is plaster, were fabricated and their acoustic characteristics were measured. Some of them showed acoustic characteristics similar to those of real bovine cancellous bone. From this study it was found that the significant acoustic characteristics of the phantom, attenuation and velocity, can be controlled by varying the concentration of flax. To some extent, the internal structure of cancellous bone was also mimicked.

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